

Motivation

Driven by a passion for understanding complex fluid dynamics and magnetic phenomena, I am committed to advancing scientific knowledge through rigorous research and collaboration. My goal is to contribute meaningfully to the scientific community while continuously growing as a researcher.

Research Interests

Fluid dynamics, mathematical modelling, magnetohydrodynamics, and dynamo theory.

Current Position

2023–Present **Postgraduate Student**, *University of Glasgow*

Conducting research on dynamo theory, focusing on force balances in magnetic field reversal regimes.

Education

2020–2021 **Master of Education**, *Allama Iqbal Open University*, Pakistan

2016–2019 **Bachelor of Education**, *Allama Iqbal Open University*, Pakistan

2014–2016 **Master of Mathematics**, *University of Punjab*, Pakistan

2010–2014 **BS (Hons)**, *Government College University Faisalabad*, Pakistan

Work Experience

2016–2022 **Secondary School Educator**, *School Education Department*, Pakistan

- Taught mathematics and physics (Grades 9–10); science (Grades 7–8).
- Served as Headmistress: led meetings, managed classes, ensured discipline.
- Responsible for IT tasks including email systems and school data management.
- Deputy Superintendent/Superintendent for national secondary examinations for 4 years.

2025–Present **Graduate Teaching Assistant (Tutor and Marker)**, *University of Glasgow*

- Tutor and marker for undergraduate mathematics courses.

Publications

- [1] **M. Sharif, A. Sarwar**, *Thermodynamics of Generalized Cosmic Chaplygin Gas Model*, **Modern Physics Letters A**, 2016.
- [2] **M. Sharif, A. Sarwar**, *Thermodynamics of Anisotropic Emergent Universe in Nonlinear Electrodynamics*, **Modern Physics Letters A**, 2016.
- [3] **A. Sarwar, R. Teed, R. Simitev**, *Dynamic force balances of bistable spherical dynamos with stress-free boundaries (manuscript in preparation)*

Conferences and Workshops

2023 UKMHD, University of Leeds, UK — Attended

2023 Scottish Fluid Mechanics Meeting, University of Strathclyde, UK — Attended

2024 Geophysical and Astrophysical Fluids and Dynamos Meeting, University of Leeds, UK — Poster: *Characteristics of Dipolar Dynamo*

- 2024 Mathematical Aspects of Geophysical and Astrophysical Fluid Dynamics, Newcastle University, UK — Attended
- 2025 Scottish Fluid Mechanics Meeting, University of Glasgow, UK — Talk: *Force Balances Characteristics of Different Dynamo Regimes*
- 2025 UKMHD, University of Exeter, UK — Talk: *Force Balances Characteristics of Different Dynamo Regimes*
- 2025 CIG, Breckenridge, Colorado, USA — Poster accepted: *Force Balances Characteristics of Different Dynamo Regimes*
- 2026 OxWIN, University of Oxford, UK — Pitch talk: Force Balances of Dipolar Dynamos
- 2026 Scottish Fluid Mechanics Meeting, University of Dundee, UK — Poster: Force balances along bistable attractor branches in simulations of rotating spherical shell dynamos
- 2026 UKMHD, University of Glasgow, UK — Poster: Force balances along bistable attractor branches in simulations of rotating spherical shell dynamos
- 2026 Rayleigh Hackathon, Fraser, Colorado, USA—collaborated with the Rayleigh developers to implement diagnostics for the curl of the momentum equation.
- 2026 EMS PhD students meeting, the Burn, Edzell—talk: Force balances along bistable attractor branches in simulations of rotating spherical shell dynamos.

--- Outreach and Public Engagement

- 2026 Volunteer at the Glasgow Science Festival, engaging with visitors and assisting with the *Sun is Full of Forces* public engagement activity.

--- Scholarships and Awards

- 2023 College of Science and Engineering PhD Scholarship, University of Glasgow
- 2022 Higher Education Commission Overseas Scholarship (Phase 3), Pakistan
- 2015 Higher Education Commission Scholarship, Pakistan
- 2014 Departmental Award, University of Punjab, Pakistan